THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of $(x+3)^2$ when $x = 7$ is: 1

(a) 106 **(b)** 58 **(c)** 79 **(d)** 100

2. 6 pipes fill a tank in 10 hours. 4 such pipes will fill it in: 1

(a) 12 hours **(b)** 16 hours

(c) 15 hours **(d)** 6.666666666666666666666666667 hours

3. The simple interest on a sum at 5% p.a. for 4 years is ₹ 600. The sum (in ₹) is: 1

(a) 3300 **(b)** 6000 **(c)** 2700 **(d)** 3000

4. Two vertical poles of heights 40 m and 22 m stand 24 m apart on level ground. The distance between their tops (in m) is: 1

(a) 30 **(b)** 42 **(c)** 28 **(d)** 31

5. A sum of ₹ 1200 buys 24 kg of rice at ₹ 50 per kg. When the price falls to ₹ 40 per kg, the same money buys (kg): 1

(a) 24 **(b)** 28 **(c)** 30 **(d)** 48

6. The value of $(x+2)(x+5)$ when $x = 4$ is: 1

(a) 44 **(b)** 54 **(c)** 26 **(d)** 58

7. A : B = 2 : 3 and B : C = 6 : 5. If A + B + C = 165, then C equals: 1

(a) 66 **(b)** 44 **(c)** 33 **(d)** 55

8. A number is increased by 25% and the result is 300. The original number is: 1

(a) 216 **(b)** 480 **(c)** 264 **(d)** 240

9. The area of a right triangle with hypotenuse 30 cm and one leg 18 cm (in cm²) is: 1

(a) 216 **(b)** 234 **(c)** 432 **(d)** 270

10. A train at 60 km/h takes 9 hours for a journey. At 45 km/h it will take (hours): 1

(a) 12 **(b)** 6.75 **(c)** 13 **(d)** 11

11. $(x+2)^2 - (x-2)^2$ equals: 1

(a) $8x$ **(b)** $2x^2 + 8$ **(c)** $4x$ **(d)** $8x + 4$

12. On a map with scale 1 : 50000, a distance of 4 cm represents: 1

(a) 200 km **(b)** 20 km **(c)** 2 km **(d)** 6 km

13. The difference between CI and SI on ₹ 5000 for 2 years at 4% p.a. (in ₹) is: 1

(a) 8 **(b)** 16 **(c)** 9 **(d)** 7

14. Which set of numbers is a Pythagorean triplet? 1

(a) 33, 44, 55 **(b)** 33, 46, 55 **(c)** 33, 44, 56 **(d)** 34, 44, 55

15. Two numbers are in the ratio 7:11 and their sum is 144. Their difference is: 1

(a) 72 **(b)** 18 **(c)** 34 **(d)** 32

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find 39×41 using $(a+b)(a-b)$. 2

17. Convert 45 km/h into m/s. 2

18. A sum of ₹ 5600 amounts to ₹ 6440 in 3 years at simple interest. Find the rate per annum. 2

19. The diagonal of a rectangle is 40 cm and one side is 24 cm. Find the other side and the area. 2

20. A hostel has food for 50 students for 16 days. How long will the food last if 10 students leave? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In how many years will ₹ 7500 amount to ₹ 9300 at 8% simple interest per annum? 3

22. Show that the triangle with sides 18 cm, 24 cm and 30 cm is right-angled and find its area. 3

23. A:B = 1:2 and B:C = 3:4. If A+B+C = 170, find the value of each. 3

24. Verify that $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$ for $a = 5$ and $b = 2$. 3

OR

The ages of a mother and daughter are in the ratio 7 : 3. The mother is 42 years old. Find their ages 4 years ago and the ratio then. 3

25. A sum of ₹ 10000 is lent for 2 years at 10% for the first year and 20% for the second year, compounded annually. Find the amount. 3

OR

The altitude drawn to the hypotenuse of a right triangle with legs 18 cm and 24 cm: find the hypotenuse and the length of this altitude. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A mixture of 60 litres has milk and water in the ratio 2:3. How much water must be added to make the ratio 2:4? 5

OR

Use identities to evaluate: 5

(i) 103^2

(ii) 88×92

(iii) $76^2 - 24^2$.

27. Anil and Bhanu start a business with ₹ 15000 and ₹ 25000. They share a profit of ₹ 800 in the ratio of their investments. Find the ratio of investments and each person's share. 5

OR

A man deposits ₹ 4000 at simple interest and another ₹ 6000 at the same rate. After 2 years the total interest is ₹ 1000. Find the rate and the interest earned on each sum. 5

END OF QUESTION PAPER **Total: 50 Marks**

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